



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
 - Data Collection API with Webscraping
 - Data Wrangling
 - Data Analysis with SQL
 - Data Visualization
 - Interactive Visual Analytics and Dashboard
 - Machine Learning Predictive Analysis
- Summary of all results
 - Success rate for launches has been increasing and SpaceX has very high rates of success with certain launches particularly from certain locations.

Introduction

- Space X competitors cost roughly \$165 million to SpaceX Falcon 9's \$60+ million figure. SpaceX will have even more of a competitive advantage with more successful launches and increased reusability.
- Using information from previous launches, we can decipher the probability of future successful launches depending on the location and type of launch as well as other factors.

Section 1

Methodology

Methodology

Executive Summary

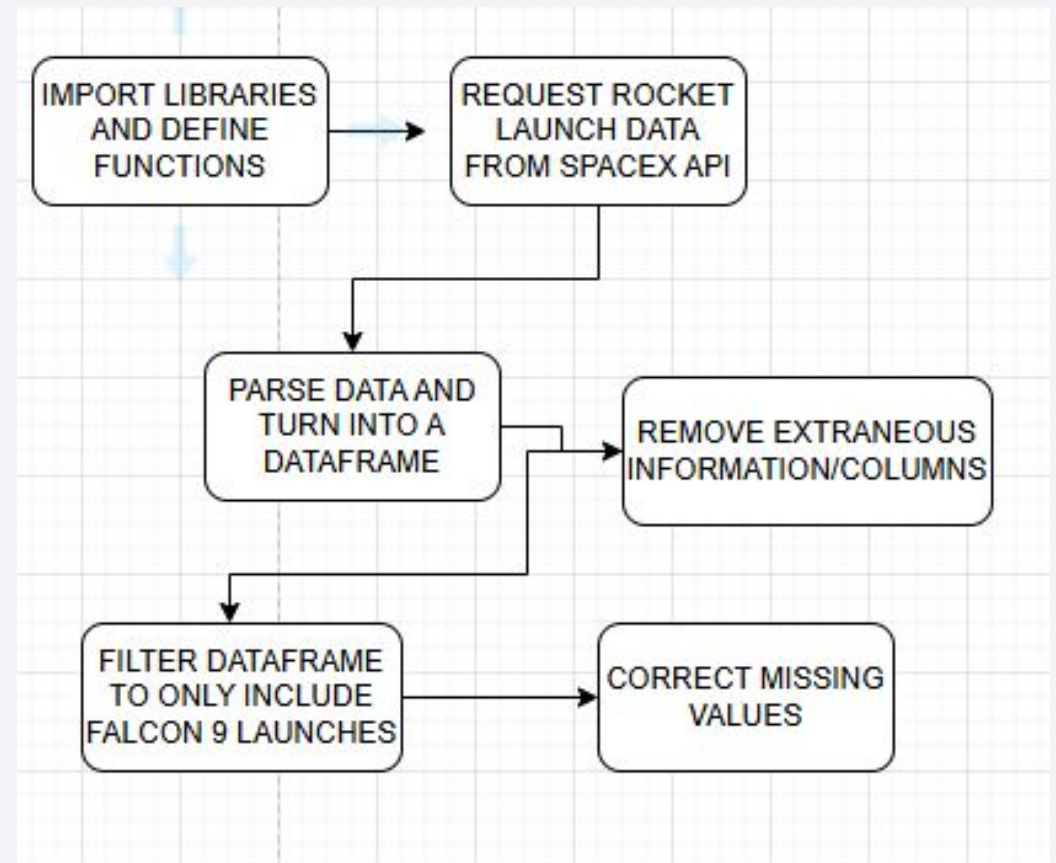
- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

- Describe how data sets were collected.
- You need to present your data collection process use key phrases and flowcharts

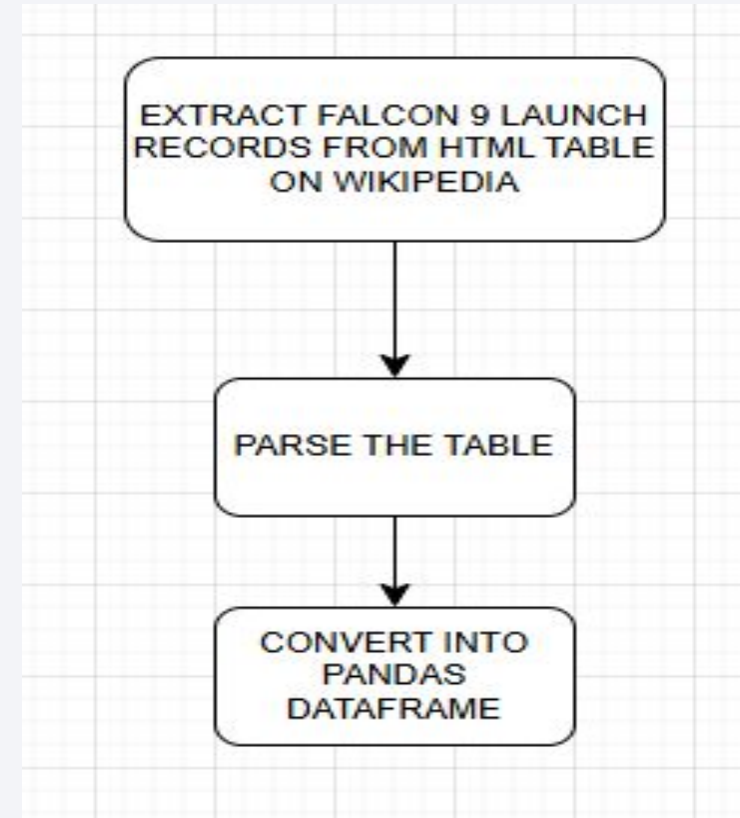
Data Collection – SpaceX API

- Data was collected from the SpaceX API as seen in the flowchart to the right.
- <https://github.com/adroit-al/Jupyter-Notebook/blob/SpaceX-Flight/jupyter-labs-spacex-data-collection-api.ipynb>



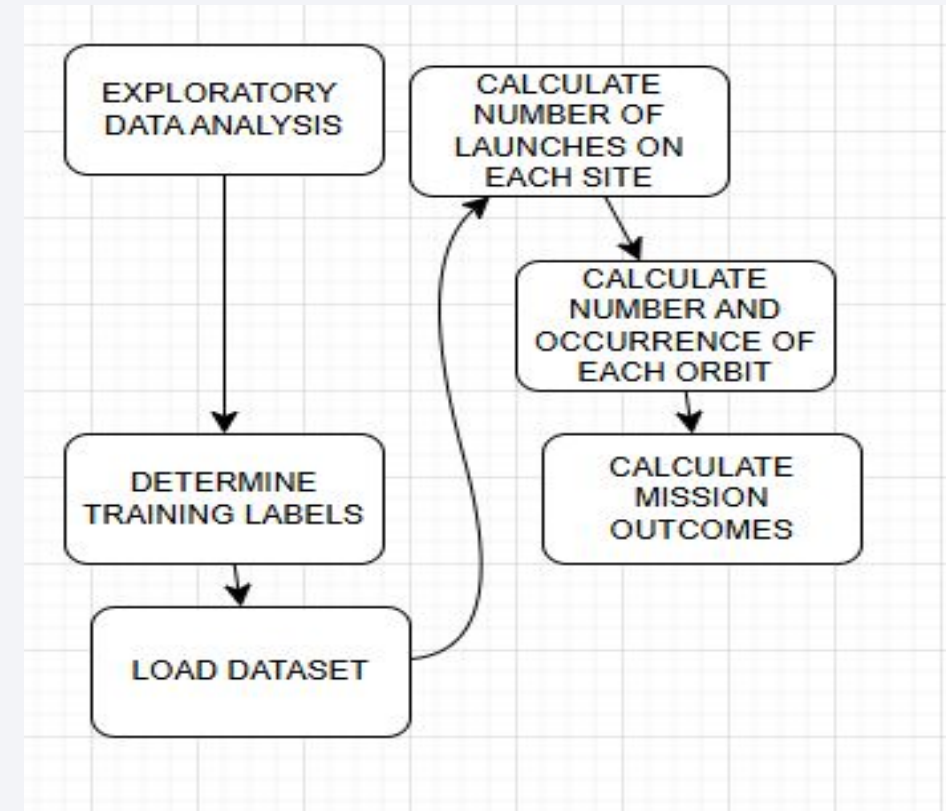
Data Collection - Scraping

- Webscraping was completed using get request from an online list of Falcon 9 launches which was then converted into a dataframe
- <https://github.com/adroit-al/Jupyter-Notebook/blob/SpaceX-Flight/jupyter-labs-webscraping.ipynb>



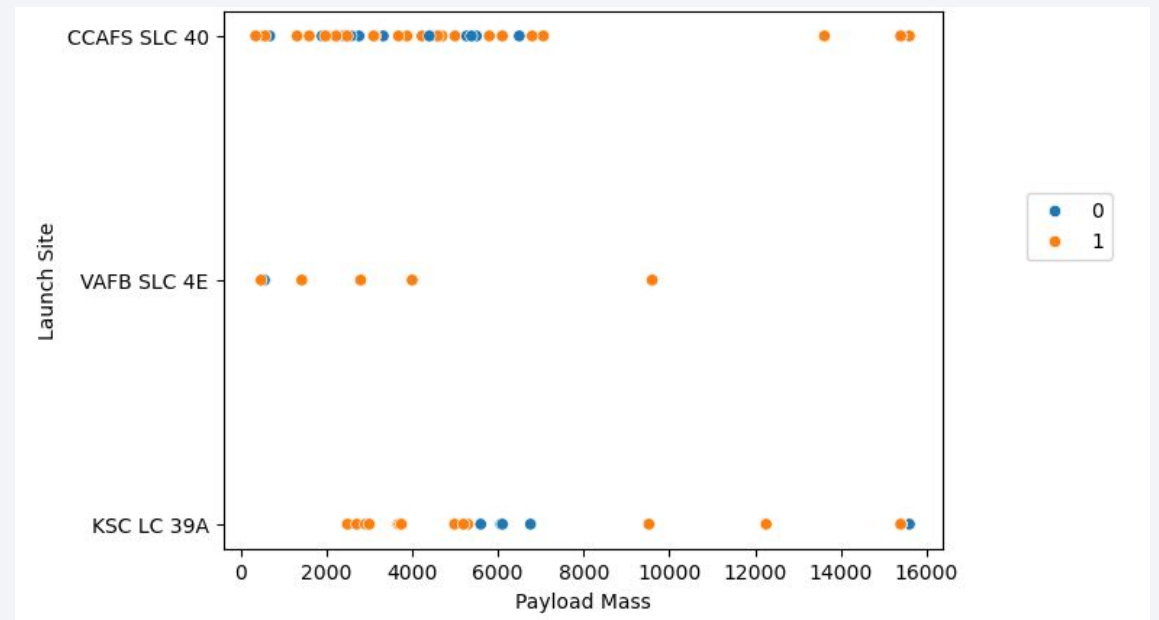
Data Wrangling

- Exploratory data analysis was conducted to determine the available information, determine missing values, and datatypes some of which were numerical while others were categorical in nature. From there, the number of launches from each location was determined as well as the result of each launch.
- <https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.dtypes.html>



EDA with Data Visualization

- Scatterplots, Bar graphs, and a lien chart were used to visualize the data, an example of which can be seen to the right.
- <https://github.com/adroit-al/Jupyter-Notebook/blob/SpaceX-Flight/edadataviz.ipynb>



EDA with SQL

- SQL queries were used to determine the following:
 - Average payload mass carried by booster version F9
 - Determine date of first successful landing outcome
 - List boosters within certain range of mass
 - List total number of successful and failure mission outcomes
 - List the month failure outcomes occurred for drone ships
 - Rank landing outcomes in specific date range
- https://github.com/adroit-al/Jupyter-Notebook/blob/SpaceX-Flight/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

- Markers, circles and lines were added to the Folium map
- The lines were to display the distance from the launch site to the coastline and the markers of varying color showed whether each launch was a success or failure
- https://github.com/adroit-al/Jupyter-Notebook/blob/SpaceX-Flight/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

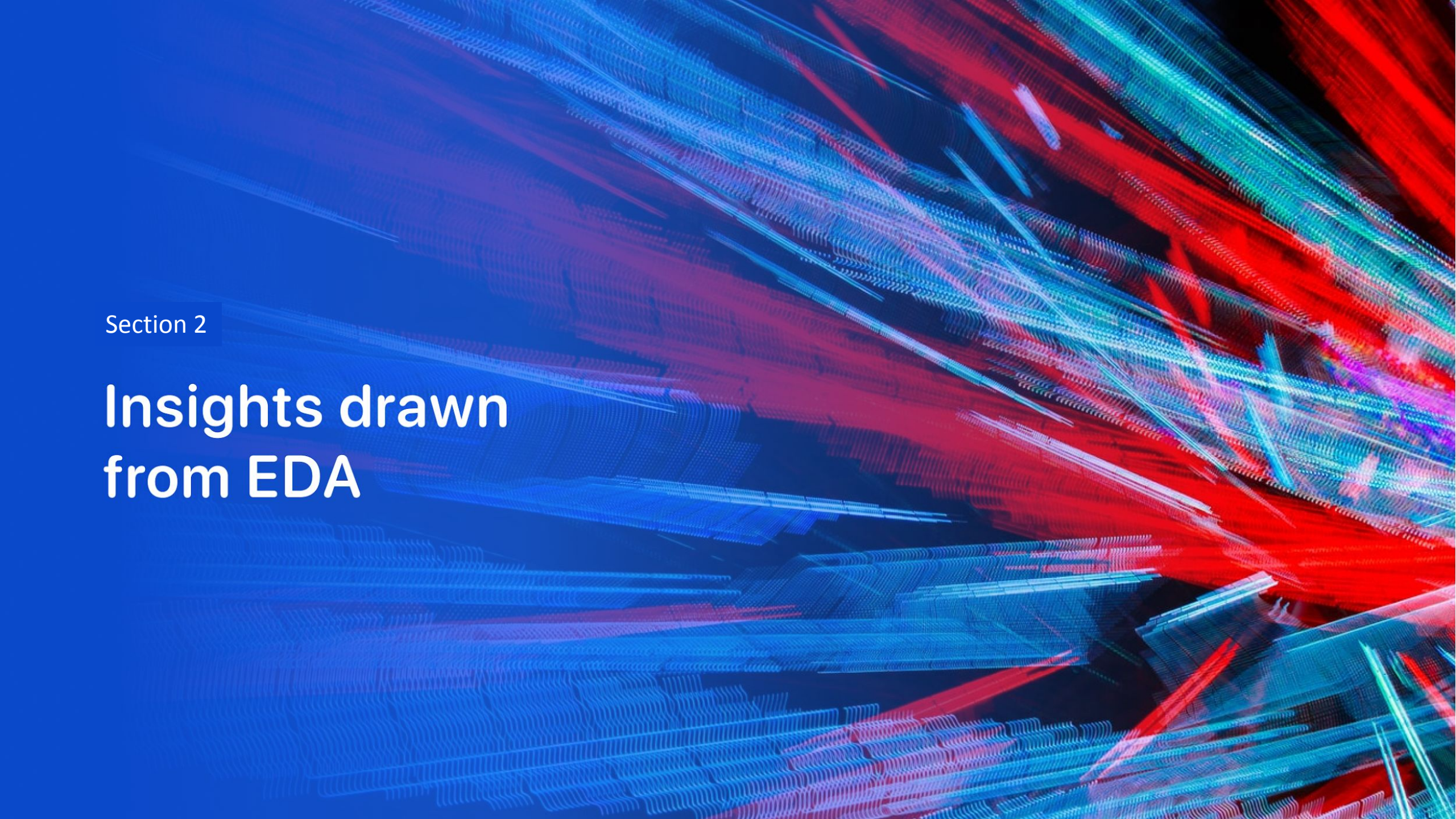
- Summarize what plots/graphs and interactions you have added to a dashboard
- The plots/graphs added include a pie chart to visualize the percentage of launches from the different locations and a scatter plot of the payload complete with launchsite details as well as whether it was a success or failure as well as input interaction with a slider so the user could select certain fields
- These plots and interactions were added to include an interactive visual so that users could easily determine the effect of payload mass and launch location on the success rate of a launch.

Predictive Analysis (Classification)

- First the data was standardized using a standard scalar and split into training and testing data. From there, different models were tested such as logregression, SVM, DecisionTree classifier, and KNN
- Different models were tested to determine which would be best to use, however, they all performed similarly. This being the case, it is best to use the model that requires the least amount of resources.
- https://github.com/adroit-al/Jupyter-Notebook/blob/SpaceX-Flight/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

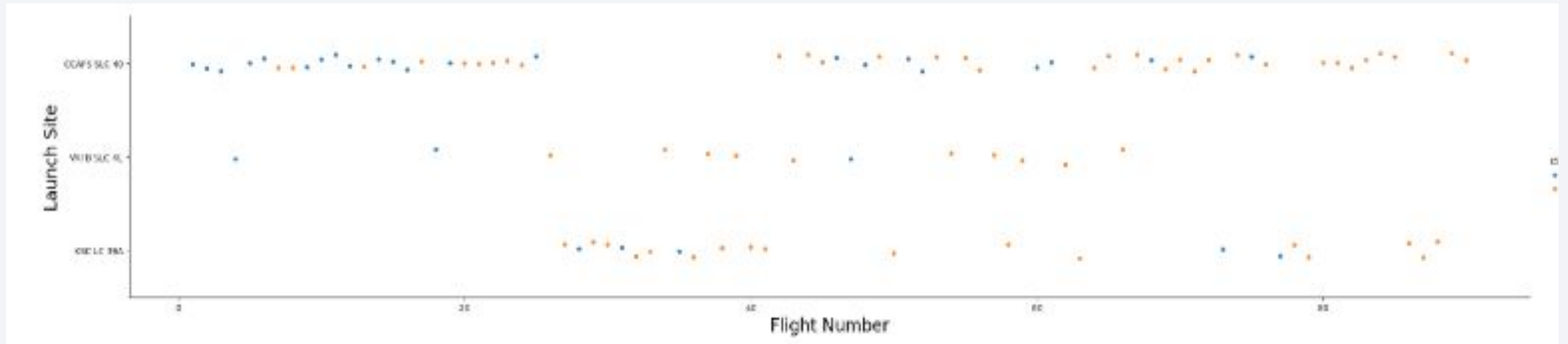
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a dynamic pattern of diagonal streaks in shades of blue and red on the right. These streaks are layered over a fine, light-colored grid, creating a sense of depth and movement, reminiscent of digital data or a complex network.

Section 2

Insights drawn from EDA

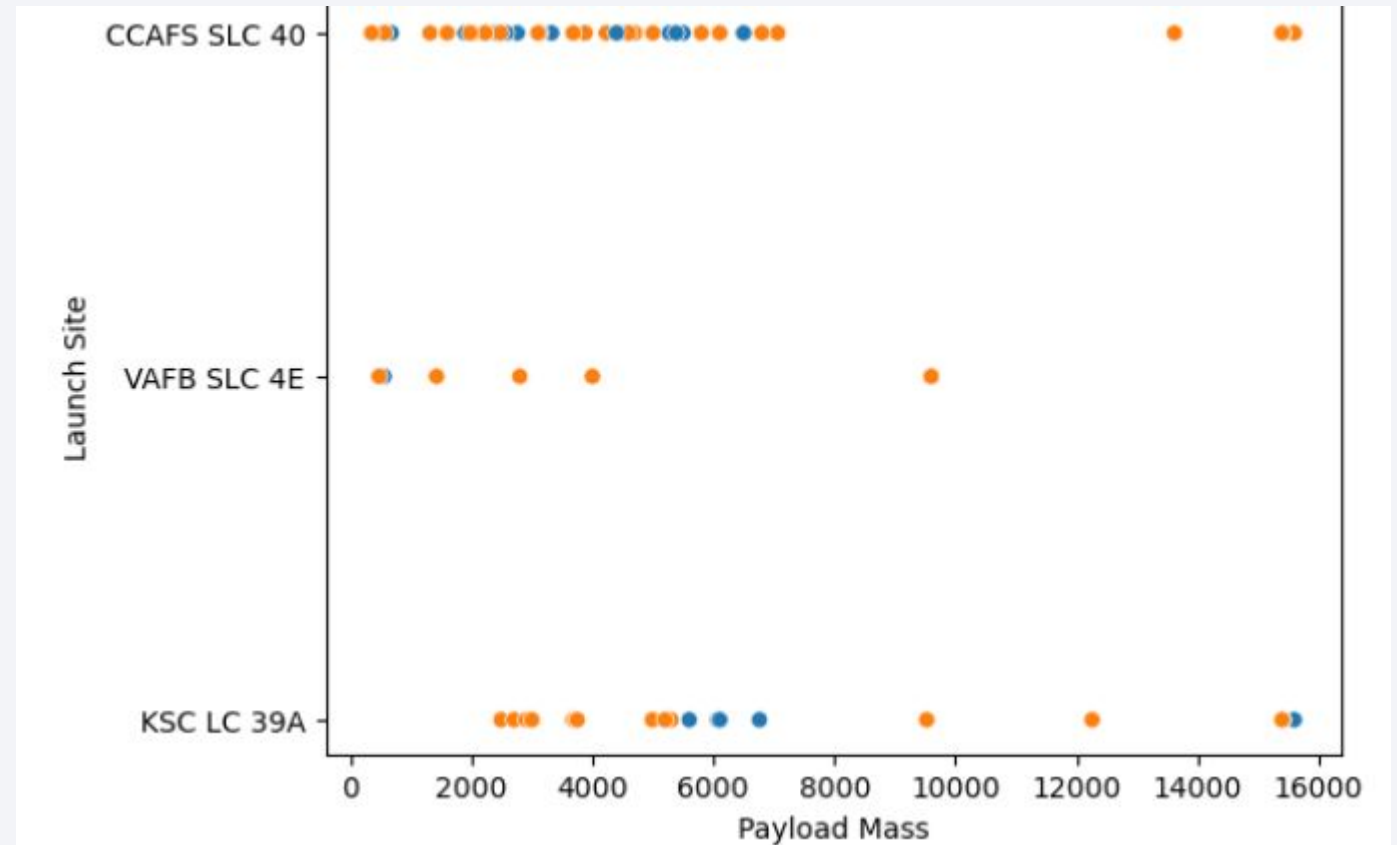
Flight Number vs. Launch Site



- Early launches resulted in many failures, however, each site's success rates improved drastically as the number of flights increased.

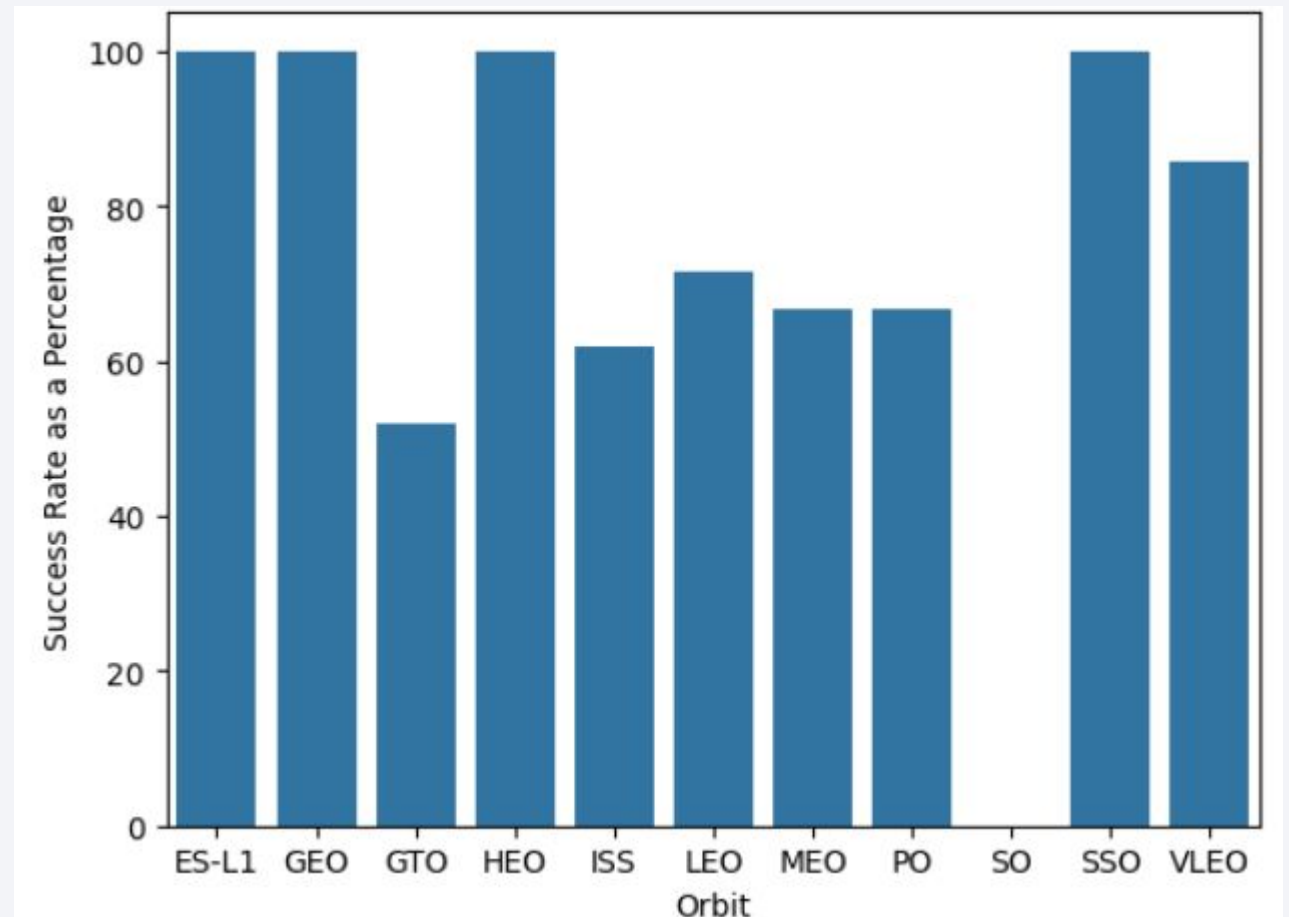
Payload vs. Launch Site

The lowest payloads had a lower chance of success with the success rate being the highest between 8,000 and 15,000kg of payload mass.



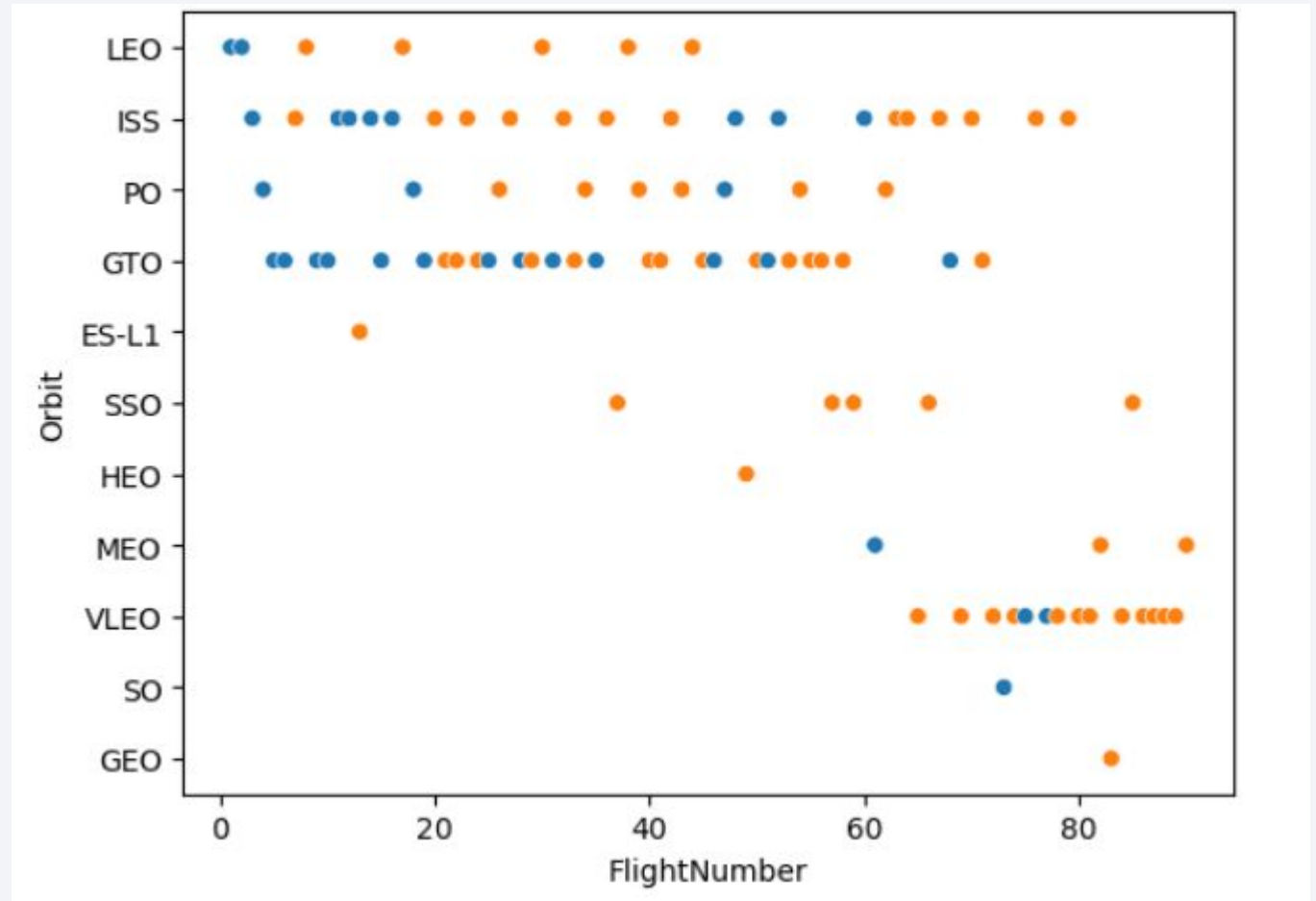
Success Rate vs. Orbit Type

- ES-L1, GEO, HEO, and SSO had the highest success rates followed by VLEO with SO having no success.



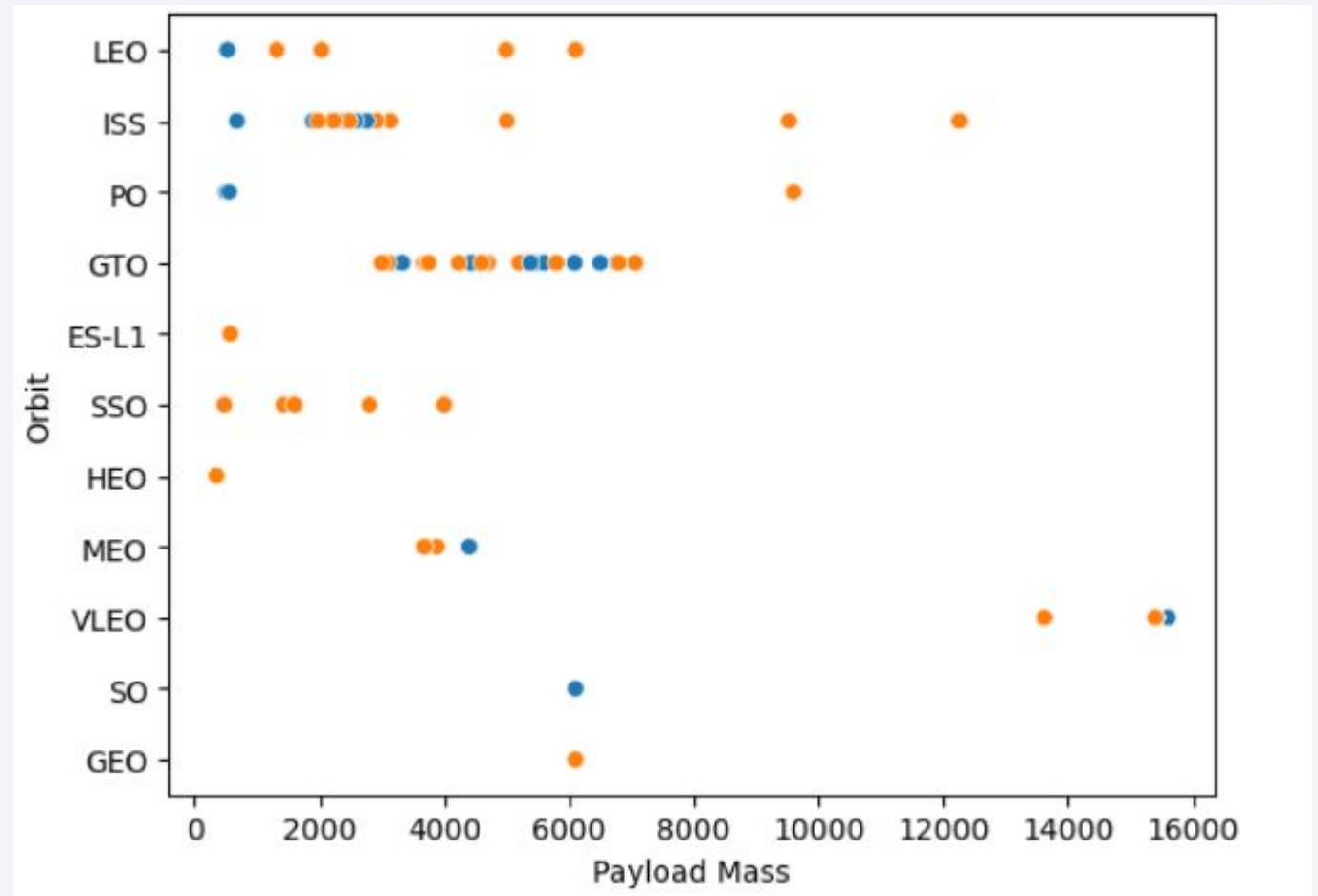
Flight Number vs. Orbit Type

- LEO greatly improved with no failures after the initial flights, VLEO also saw relatively few failures.
- SSO has remained significantly successful and ES-L1 has a 100% success rate but it should be noted that with more launches, the success rate will likely decrease.
- Overall, there is a trend of increasing success rates as the flight number increases.



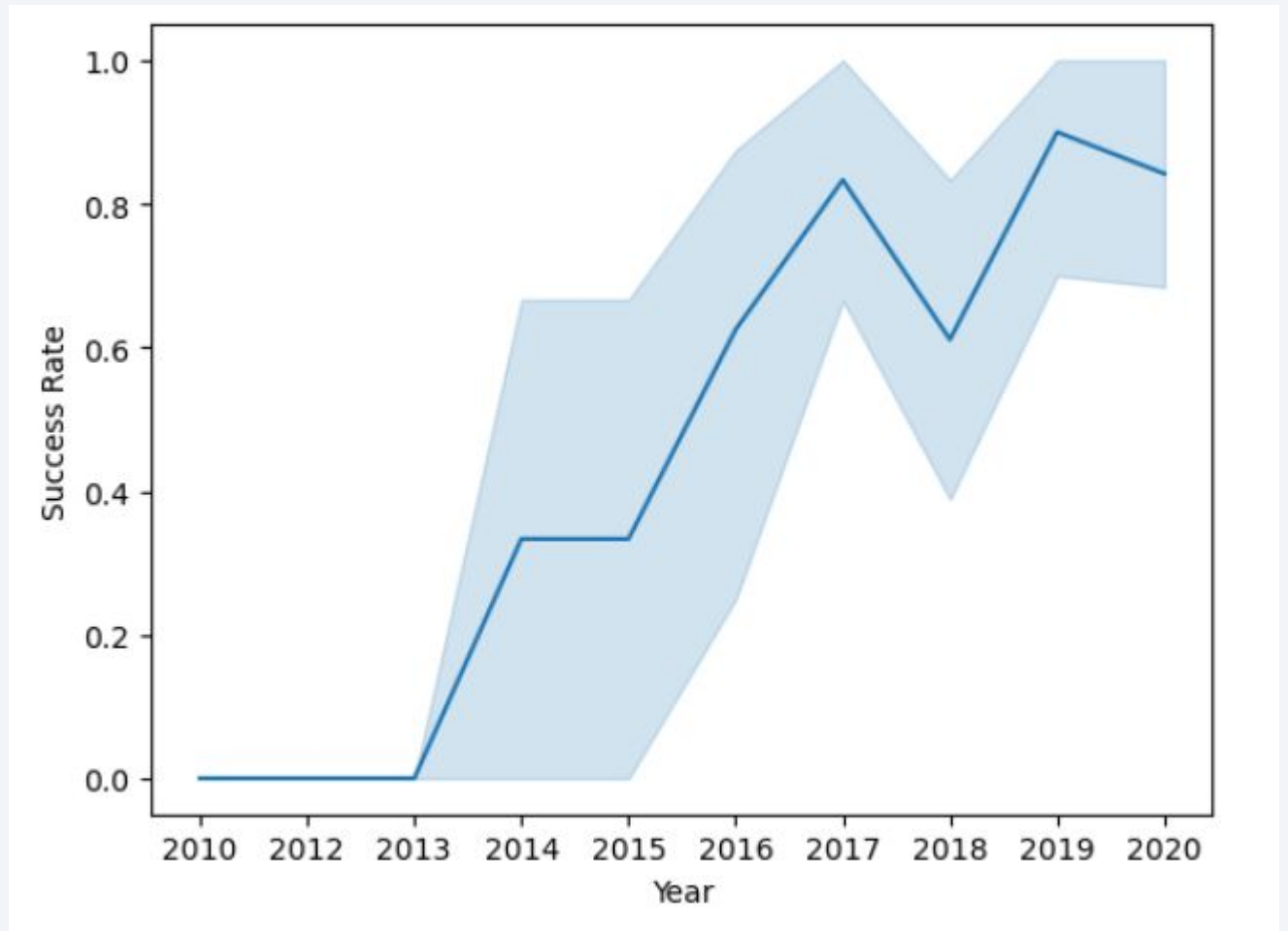
Payload vs. Orbit Type

- It is unclear how significant payload mass is on the success or failure of a flight. This particularly comes into focus looking at GTO launches.



Launch Success Yearly Trend

- The success rate greatly increased from 2013 to 2017 with more failures in 2018 and beyond with peaks in 2018 and 2019 around 80% and 90% success rates, respectively.



All Launch Site Names

- The unique launch sites are as seen below based on the SQL query

```
In [13]: %sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[13]: Launch_Site
```

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- See 5 Launch Sites below obtained in a SQL query

```
In [16]: %sql SELECT Launch_Site FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%'
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[16]: Launch_Site
```

```
CCAFS LC-40
```

```
CCAFS LC-40
```

```
CCAFS LC-40
```

```
CCAFS LC-40
```

```
CCAFS LC-40
```

Total Payload Mass

- See the sum of payload mass below which was the result of the below query

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
SUM(PAYLOAD_MASS__KG_)
```

```
619967
```

Average Payload Mass by F9 v1.1

- See the average payload mass from the result of the query below.

```
%sql SELECT AVG(PAYLOAD_MASS_KG_) FROM SPACEXTABLE
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
AVG(PAYLOAD_MASS_KG_)
```

```
6138.287128712871
```

First Successful Ground Landing Date

- The date was obtained using the query below.

```
%sql SELECT MIN(DATE) FROM 'SPACEXTBL' WHERE "Landing _Outcome" = "Success (ground pad)";
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
MIN(DATE)
```

```
01-05-2017
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- Successfully landed drone ships having a payload mass greater than 4000 but less than 6000 are shown using the query below.

```
%sql SELECT DISTINCT Booster_Version, PAYLOAD FROM SPACEXTABLE WHERE LANDING_OUTCOME LIKE "Success"
```

* sqlite:///my_data1.db
Done.

Booster_Version	Payload
F9 FT B1022	JCSAT-14
F9 FT B1026	JCSAT-16
F9 FT B1021.2	SES-10
F9 FT B1031.2	SES-11 / EchoStar 105

Total Number of Successful and Failure Mission Outcomes

- Outcomes were obtained using the query below.

List the total number of successful and failure mission outcomes

```
%sql SELECT Mission_Outcome, COUNT(Mission_Outcome) FROM SPACEXTABLE GROUP BY Mission_Outcome
```

```
* sqlite:///my_data1.db
```

Done.

Mission_Outcome	COUNT(Mission_Outcome)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- Below are the boosters which have carried the maximum payloads.

Booster_Version	Payload	PAYLOAD_MASS_KG_
F9 B5 B1048.4	Starlink 1 v1.0, SpaceX CRS-19	15600
F9 B5 B1049.4	Starlink 2 v1.0, Crew Dragon in-flight abort test	15600
F9 B5 B1051.3	Starlink 3 v1.0, Starlink 4 v1.0	15600
F9 B5 B1056.4	Starlink 4 v1.0, SpaceX CRS-20	15600
F9 B5 B1048.5	Starlink 5 v1.0, Starlink 6 v1.0	15600
F9 B5 B1051.4	Starlink 6 v1.0, Crew Dragon Demo-2	15600
F9 B5 B1049.5	Starlink 7 v1.0, Starlink 8 v1.0	15600
F9 B5 B1060.2	Starlink 11 v1.0, Starlink 12 v1.0	15600
F9 B5 B1058.3	Starlink 12 v1.0, Starlink 13 v1.0	15600
F9 B5 B1051.6	Starlink 13 v1.0, Starlink 14 v1.0	15600
F9 B5 B1060.3	Starlink 14 v1.0, GPS III-04	15600
F9 B5 B1049.7	Starlink 15 v1.0, SpaceX CRS-21	15600

2015 Launch Records

- Failed landing outcomes in drone ships as well as there booster versions and launch sites for the year 2015 can be seen below.

Month	Year	Landing_Outcome	Booster_Version	Launch_Site
01	1	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	1	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The following are the results in descending order by date of landing outcomes from 2017/03/20 to 2010/06/04

Landing_Outcome	Date
No attempt	2017-03-16
Success (ground pad)	2017-02-19
Success (drone ship)	2017-01-14
Success (drone ship)	2016-08-14
Success (ground pad)	2016-07-18
Failure (drone ship)	2016-06-15
Success (drone ship)	2016-05-27
Success (drone ship)	2016-05-06
Success (drone ship)	2016-04-08

Failure (drone ship)	2016-03-04
Failure (drone ship)	2016-01-17
Success (ground pad)	2015-12-22
Precluded (drone ship)	2015-06-28
No attempt	2015-04-27
Failure (drone ship)	2015-04-14
No attempt	2015-03-02
Controlled (ocean)	2015-02-11
Failure (drone ship)	2015-01-10

Uncontrolled (ocean)	2014-09-21
No attempt	2014-09-07
No attempt	2014-08-05
Controlled (ocean)	2014-07-14
Controlled (ocean)	2014-04-18
No attempt	2014-01-06
No attempt	2013-12-03
Uncontrolled (ocean)	2013-09-29
No attempt	2013-03-01

No attempt	2012-10-08
No attempt	2012-05-22
Failure (parachute)	2010-12-08
Failure (parachute)	2010-06-04

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a curved line separating the dark surface from the deep blue of space.

Section 3

Launch Sites Proximities Analysis

Launch Sites on Global Map

As seen from the map, the launch sites are located near the coastline and close to the equator.

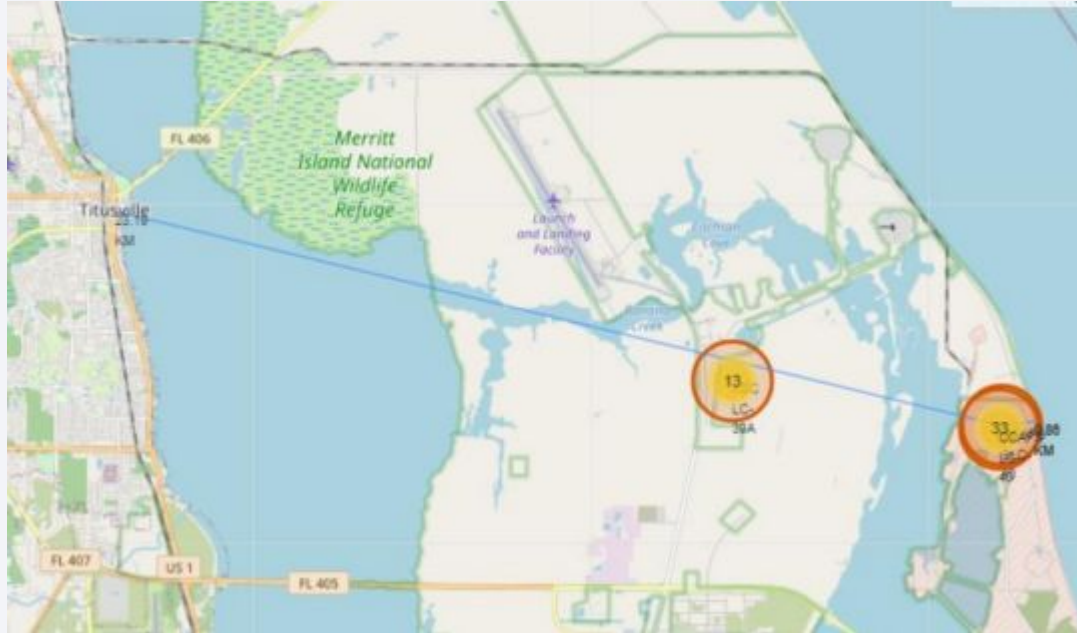


Launchsite Map and Outcomes

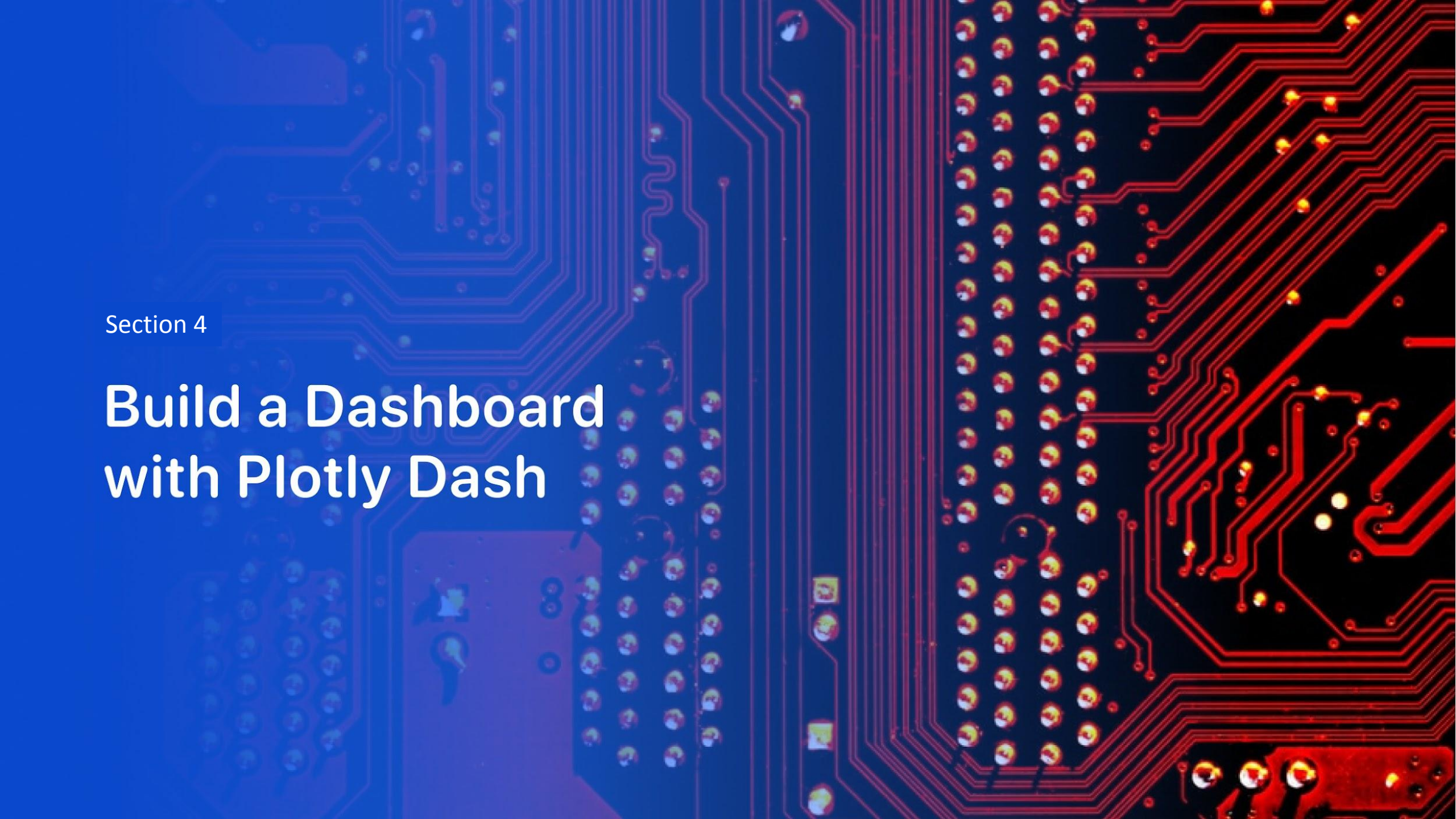
- Launch Sites as well as outcomes. Success was much more likely in KSC LC-39A than other launch sites.



Launchsite and Proximities Map



- Distance between launchsite and its proximities.



Section 4

Build a Dashboard with Plotly Dash

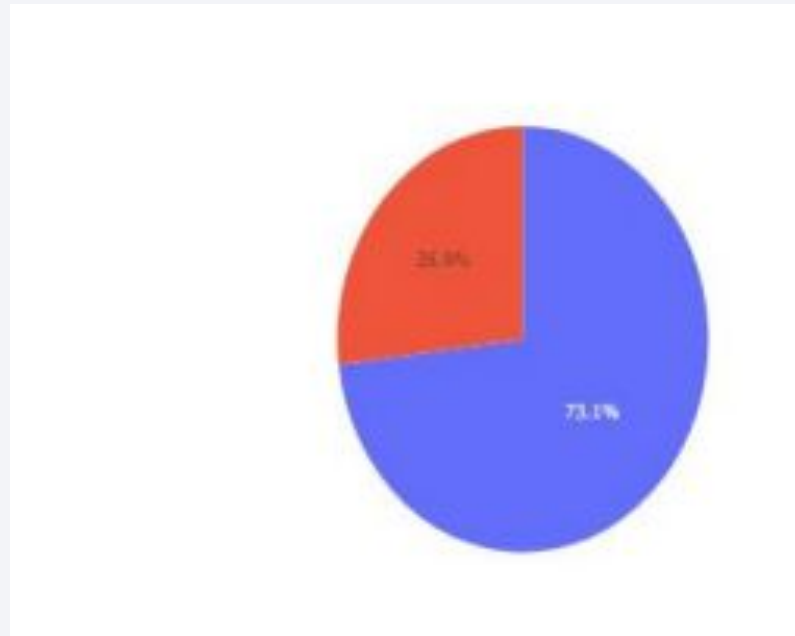
Successful Counts for Launch Sites

The launchsite with the most successful counts was KSC LC-39A followed by CCAFS LC-40

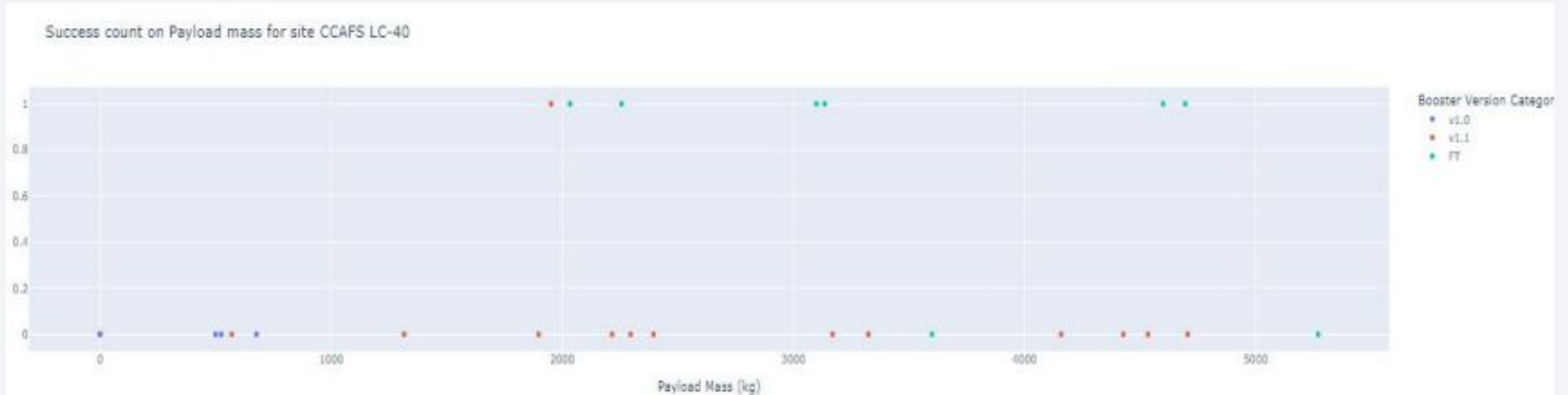


Success Rate KSC LC-39A

- KSC LC-39A had the highest success rate with over 75% of launches resulting in success.



Launch Outcome by Payload



- This shows the effect that payload mass and booster version had on success rate.

Section 5

Predictive Analysis (Classification)

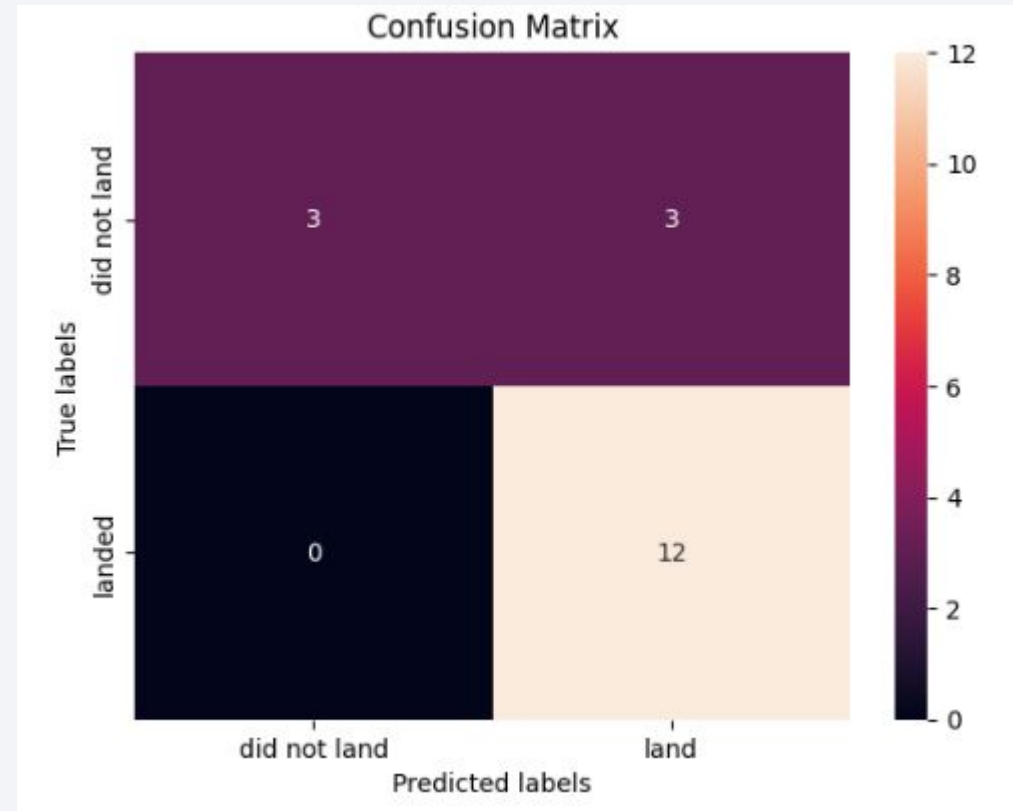
Classification Accuracy

- Decision Tree was the most accurate among the classifiers.



Confusion Matrix

There were no false negatives and overall the results were fairly reliable for most scenarios.



Conclusions

- Some launch sites saw greater success than others.
- The data was not sufficient to determine the reason for success or failure
- ES-L1, HEO, SSO, and GEO were the most successful orbits
- In general, the company saw increasing success as the flight number increased
- Some of the locations had a low success rate but many of their launches were earlier. There were likely other factors besides location that caused these lower success rates.

Thank you!

